Technological Advancements in the Medical Field

A glimpse into the future

The health care system is one of the most important pillars of any country. It is an engine for innovation that develops and provides advanced, life-enhancing treatments and offers a wide set of choices for consumers of healthcare. One of India's largest sectors both in terms of revenue and employment, healthcare comprises hospitals, medical devices, clinical trials, outsourcing, telemedicine, health insurance and medical equipment. The Indian healthcare sector is growing at a rapid pace due to its strengthening coverage, services and increasing expenditure by public as well as private players. The medical field has seen exponential advancements in recent years and this National Doctor's Day is a perfect opportunity to look at some of these technological upgrades in the medical field. The medical field has come a long way since the invention of eyeglasses and the stethoscope. The far-reaching availability of the Internet, the expansion of a more affluent middle class, and an aging global population are all driving change in the healthcare industry. More than a billion people will need reskilling in medical technology by 2030. New technologies in medicine need to be used together, and integrated attempts to do so already exist. The COVID-19 pandemic forced healthcare into the future resulting in several upcoming medical technologies being tested on a global scale. The challenge is how those technologies can be used together in a post-pandemic world.

Artificial intelligence takes on many different forms in healthcare. The primary trend for AI in healthcare 2022 will be in utilizing machine learning to evaluate large amounts of patient data and other information.

Innovation has always been a cornerstone of health care, and many specialties are evolving and benefiting from technology and medical advances. Today technologies like Artificial intelligence, Virtual reality, Neurotechnology, 3D printing, Telemedicine, mRNA technology, immunotherapy, Blockchain technology and Fluid biopsy in cancer treatment are some of the latest innovations being used for treating patients. The medical technology industry in India is one of the largest in Asia growing at 10-12% per year until 2025. Globally the medical technology market is expected to be $600 billion by 2025 implying a manufacturing base of $200 billion by then. India is expected to capture 10% of that share by 2025. The Indian medical device market consists of medical instruments, orthopedic and ophthalmic devices, scanning devices, bandages, syringes, needles, and catheters. The application of future digital technologies in the health sector can lead to accessible, affordable, and quality health care. Also, it can reduce human involvement in risky functions. Human involvement in diagnosis, treatment, and hospital care of infected patients puts them at greater risk of contracting the disease. For instance, many frontline warriors fighting COVID-19 have lost their lives. In this context, technologies such as artificial intelligence (AI), autonomous systems, blockchain, cloud and quantum computing, data analytics, and 5G can help in addressing the issue. The new technologies can improve the welfare of societies and reduce the impact of communicable diseases. Further, it can reduce the chances of hospital staff contracting the infection. Advancements in surgical devices and equipment have constituted an integral part of Indian medicine innovation. The emergence of a range of precision surgical tools for different spheres of care such as gynecology, wound closure, cardiovascular, orthopedic, neurosurgery, plastic and reconstructive surgery. Diagnostics has received a remarkable impetus due to Covid-19. With improvements in molecular diagnostics, there has been a definitive shift towards home collection with emphasis on precision testing, automated analysis and shorter turnarounds backed by application-based coordination and logistics systems. There have also been technological developments for non-COVID diseases such as cancer, HIV and TB and several others. While the Indian Council of Medical Research has developed a battery-operated portable device named TrueNat for the diagnosis of TB, a private firm has brought down sample analysis time for TB by one-tenth.

